

Mehrnoosh Emamifar

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Professional Summary

Machine learning expert with 8 years of academic and 3 years of industry experience. Experienced in developing machine learning and deep-learning-based models and agentic workflows for biological, healthcare, clinical and EHR data. Collaborative and multidisciplinary researcher with 6 peer-reviewed publications, including 4 first-authored and 2 in Nature Communications and Nature Aging. Seeking to leverage data science and machine learning skills to solve challenging life science problems in the pharmaceutical sector.

Professional Research Experiences

Graduate Research Assistant, Northeastern University Boston, MA | Jan 2024 - Present

- Developing machine learning and deep learning models including CNN, LSTM, attention-based classifier, autoencoders, contrastive learning and PCA in enumeration of circulating tumor cells (CTCs) during cancer metastasis using an optical device named Diffuse in-vivo Flow cytometry (DiFC).
- Reduced false detections by a factor of 5 compared to our previous signal processing approach, and achieved sensitivity, specificity, and AUC of 93%, 96%, and 98%, respectively.
- Performed image analysis and whole-cell segmentation of tissue multiplex immunofluorescence images using the deep learning-based method Mesmer.

Data Science Intern, AstraZeneca, Oncology Data Science Department Waltham, MA | Jun 2025 - Aug 2025

- Developed an AI agent equipped with a range of tools for flow cytometry data analysis, using OpenAI LLMs, Chainlit, Streamlit and Pydantic AI.
- Analyzed longitudinal patient data and implemented machine learning models to classify patients as responders or non-responders to immunotherapy treatments.
- Presented work to 100+ global attendees at Oncology Data Science (ODS) team meeting.

Bioinformatics Collaborator (Voluntarily), Harvard Medical School Boston, MA | Sep 2023 - Jan 2024

- Applied data science techniques to analyze and visualize epigenetic and transcriptomic age-dependent changes in blood across multiple cohorts, identifying multi-omics aging-related genes.
- Contributed to development of [Biolearn](#), an open-source toolkit for analysis of biomarkers of aging and age-related diseases.
- Led to two papers being published in *Nature Aging* and *Nature Communications*.

Graduate Research Assistant, Northeastern University Boston, MA | Sep 2022 - Jan 2024

- Developed physics-informed reinforcement learning and Bayesian models to enhance the action control system of autonomous vehicles at signalized intersections.
- Developed and assessed different policies for action selection under temporally sparse partial observations.
- Produced two conference and one journal papers.

Data Science Intern, Tali-AI Toronto, Canada (Remote) | Apr 2022 - Aug 2022

- Data Preprocessing and finetuning BERT-based LLMs.
- Contributed to the improvement of an LLM-based virtual assistant for physicians, designed for medical scribing, transcription, and question-answering tasks, integrating with EHR.

Data Scientist, Tapsi Tehran | Sep 2018-Sep 2021

- Applied machine learning models including timeseries analysis methods, CNN, autoencoder, PCA and clustering to travel time prediction in ride hailing services.
- Used PySpark to process and filter hundreds of GBs of Parquet data from GPS records across thousands of drivers.
- Achieved 11% Improvement in the MAE of short-term traffic prediction and 8.3% enhancement in the MAPE of travel time prediction in congested time intervals compared to the method in Tapsi.

Education

Northeastern University – Ph.D. Bioengineering	Boston, MA Expected Dec 2026
University of Tehran – M.Sc. Artificial Intelligence and Robotics	Tehran Sep 2021
University of Tehran – B.Sc. Computer Engineering	Tehran Jul 2018

Skills

- **Machine Learning and Data Analysis:** Deep Learning, Large Language Model (LLM), Agentic AI, Convolutional Neural Network (CNN), Time Series Analysis, Dimension Reduction, Reinforcement Learning
- **Programming and Tools:** Python and Related Libraries (e.g. TensorFlow, NumPy, Pandas, SciPy, Scikit-Learn, ScanPy, Pydantic AI), Chainlit, Streamlit, Git, Spark, Cloud Computing (AWS, Domino Data Lab)
- **Soft Skills:** Analyze, Research, Plan and Organize, Counsel and Serve, Interpersonal Relations

Talks and Posters

- New England Science Symposium (NESS), Harvard Medical School, Boston, MA, 2026.
- PhD Research Expo, Northeastern University, Boston, MA 2023 and 2026.
- Boston Early Career Symposium, AstraZeneca, Waltham, MA, 2025.
- Optica Biophotonics Congress, Coronado, CA, 2025.
- Northeast Symposium on Biomedical Optics (NESBO), Massachusetts General Hospital, Boston, MA, 2024 and 2025.
- Gordon Research Conference (GRC) and Seminar (GRS) on Optics and Photonics in Medicine and Biology, Lewiston, ME, 2024.

Publications

- M. Emamifar, J. Lee, J. Pace, C. Bellini, M. Niedre, "[Machine Learning Approach for Signal Processing in Diffuse in-vivo Flow Cytometry](#)", Under Review, "Biophotonics Discovery", 2026.
- M. Emamifar, J. Lee, J. Pace, C. Bellini, M. Niedre, "[Machine Learning Approach for Signal Processing in Diffuse in-vivo Flow Cytometry](#)", Bio-Optics: Design and Application 2025.
- K. Ying, S. Paulson, M. Perez-Guevara, M. Emamifar, et. al., "[A Unified Framework for Systematic Curation and Evaluation of Aging Biomarkers](#)", Nature Aging, 2025.
- M. Moqri, J. Poganik, C. Herzog, K. Ying, Q. Chen, M. Emamifar, et. al., "[Integrative epigenetics and transcriptomics identify aging genes in human blood](#)", Nature Communication, 2024.
- M. Emamifar, SF. Ghoreishi, "[Physics-Informed Particle-Based Reinforcement Learning for Autonomy in Signalized Intersections](#)", International Journal of Intelligent Transportation Systems Research, 2024.
- M. Emamifar, SF. Ghoreishi, "[Uncertainty-Aware Reinforcement Learning for Safe Control of Autonomous Vehicles in Signalized Intersections](#)", IEEE Conference on Artificial Intelligence (CAI), 2023.
- M. Emamifar, SF. Ghoreishi, "[Modeling and State Estimation of Autonomous Vehicles in Signalized Intersections](#)", International Conference on Transportation and Development (ICTD), 2023.

Honors and Awards

- PhD Network Travel Award, Northeastern, 2024 and 2025.
- Bioengineering Department Travel Award, Northeastern, 2024 and 2025.
- Winner of the 3-Minute Thesis Competition hosted by Graduate Women in Science and Engineering (GWISE), Northeastern University, 2025.
- Ranked among the top 1% nationwide in entrance examinations for B.Sc. and M.Sc. programs, Tehran, 2013, 2018.

Certifications

- [Data Science Math Skills](#) - Duke University, 2020.
- [Getting Started with AWS Machine Learning](#) - Amazon Web Services (AWS), 2020.
- [Developing AI Applications on Azure](#) - LearnQuest, 2020.
- [Teamwork Skills: Communicating Effectively in Groups](#) - University of Colorado Boulder, 2020.